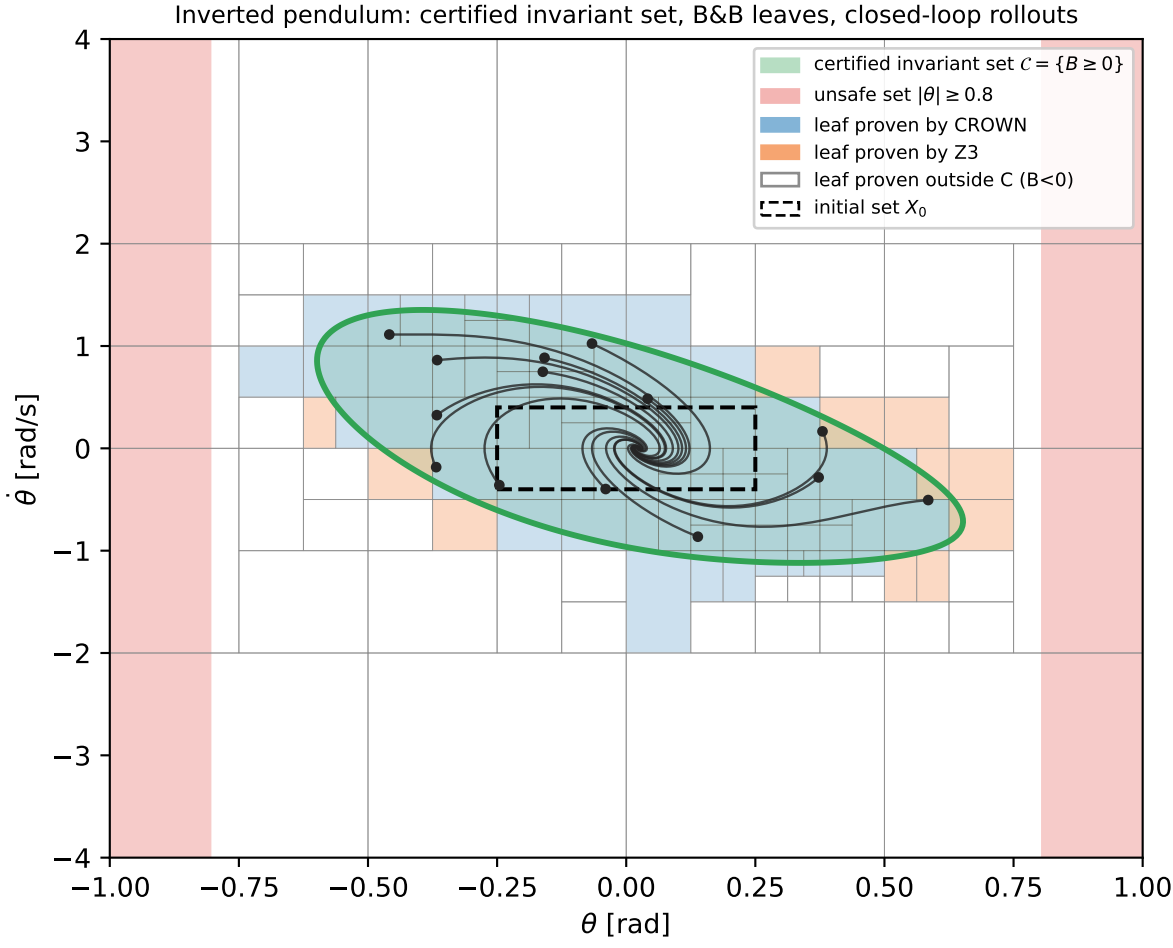
Inverted pendulum: certified invariant set, B&B leaves, closed-loop rollouts
certified invariant set $\mathcal{C} = \{B \geq 0\}$
unsafe set $|\theta| \geq 0.8$
leaf proven by CROWN
leaf proven by Z3
leaf proven outside C (B<0)
initial set X_0
$\dot{\theta}$ [rad/s]
θ [rad]
4
3
2
1
0
−1
−2
−3
−4
−1.00
−0.75
−0.50
−0.25
0.00
0.25
0.50
0.75
1.00